

Defocused transfer function for a partially coherent microscope and application to phase retrieval

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The defocused weak-object transfer function of a partially coherent bright-field microscope is calculated. For weak defocus, this can be expressed analytically. Use of this transfer function for phase restoration (quantitative phase retrieval) from images of weak mixed phase-amplitude objects is discussed. © 2004 Optical Society of America

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1. INTRODUCTION

The transfer function is a useful concept for investigating the imaging properties of a spatially invariant imaging system. It can be used to compare the performance of different systems, to calculate the image of an arbitrary object structure, and as the basis of image restoration. The optical transfer function for a defocused, paraxial, incoherent system with an aberration-free circular pupil was given by Hopkins.¹ A bright-field microscope in general behaves as a partially coherent system,² in which the coherence parameter S (which is more correctly called the incoherence parameter) is given by the ratio of the numerical aperture of the condenser to that of the objective:

$$S = \frac{n_c \sin \alpha_c}{n_o \sin \alpha_o}, \quad (1)$$

where $\alpha_{c,o}$ are the semiangular apertures of the condenser and objective lenses, respectively, and $n_{c,o}$ are the refractive indices of the immersion media of the condenser and objective lenses. If $S = 0$, the system behaves coherently, that is, it is linear in amplitude, whereas if $S \rightarrow \infty$ it behaves incoherently, that is, it is linear in intensity. Other cases, including the special case when $S = 1$, called complete, full, or matched illumination, behave in a partially coherent manner. They are not linear in either amplitude or intensity but are bilinear, meaning that the image intensity is a functional of the object amplitude at two points. Imaging can then be expressed in terms of a partially coherent transmission cross coefficient² $C(\mathbf{m}; \mathbf{p})$, where $\mathbf{m}, \mathbf{p}$ are the two-dimensional grating vectors of two spatial frequencies present in the object amplitude. For a weak object, the diffracted light is weak compared with the nondiffracted light, so that interference of diffracted light with diffracted light can be neglected, and the system again becomes linear and can be described in terms of a weak-object transfer function (WOTF) $C_W(\mathbf{m}) = C(\mathbf{m}; \mathbf{0})$. The WOTF is in general complex, the real and imaginary parts imaging the real and imaginary parts of the object amplitude into the image intensity. Until the introduc-

tion of the Zernike phase-contrast technique,³ defocus was widely used (since about 1900) as a method of imaging weak phase objects, such as unstained biological specimens, in a bright-field microscope. "Every microscopist knows that transparent objects show light or dark contours under the microscope in different ways varying with defocus..." (Ref. 3, p. 686). It is also well known that the amplitude and phase of a weak object can be extracted from a pair of defocused images. The defocused WOTF has been presented previously, but based on a slit-pupil model.⁴

The generalization to the three-dimensional transfer function was introduced by Frieden⁵ for the incoherent case and by Streibl⁶ for the WOTF of a partially coherent system. Streibl⁶ showed how phase and amplitude information in a weak object are imaged by the real and imaginary parts of the WOTF. The two-dimensional defocused WOTF can be calculated by a one-dimensional Fourier transformation of the three-dimensional WOTF.^{7,8} However, it can also be calculated directly, with a method similar to that used to calculate the defocused OTF for an incoherent system, thus giving an integral expression for the defocused WOTF, which can be expressed analytically for the case of weak defocus. This is particularly useful for analysis of the method of phase imaging based on the transport equation of intensity.⁹⁻¹¹ The cited papers show that the phase of the image field can be extracted quantitatively from defocused images. Streibl¹⁰ recognized that it is the phase of the image, rather than the phase of the object, that is thus obtained. He then suggested that "if the point spread function of the imaging system is known one might calculate back from the image plane to the object" (p. 7). However, he does not describe how this might be done for a strong phase object in a partially coherent system.

2. WEAK-OBJECT TRANSFER FUNCTION FOR AN IN-FOCUS BRIGHT-FIELD MICROSCOPE

Hopkins² has described how the partially coherent transmission cross coefficient for a bright-field microscope can

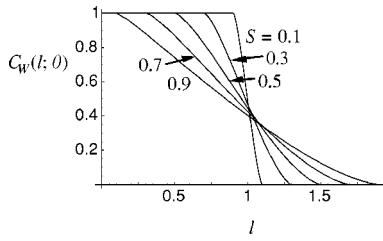


Fig. 1. In-focus WOTF for a partially coherent bright-field microscope with coherence parameter S .

be calculated as the area of overlap of three circles expressed as functions of a displacement vector ξ : the condenser pupil, and two objective pupils displaced by the spatial frequency vectors $\mathbf{m}$ and $\mathbf{p}$, respectively. In general, the weighted area over $|P_{\text{condenser}}(\xi)|^2$, $P(\xi - \mathbf{m})$, and $P^*(\xi - \mathbf{p})$ must be calculated, where $P_{\text{condenser}}$ and P are the condenser and the objective pupil functions, respectively. The WOTF is then given by the weighted area of overlap of $|P_{\text{condenser}}(\xi)|^2 P^*(\xi)$ and $P(\xi - \mathbf{m})$. For the usual case when $0 \leq S \leq 1$, for an in-focus system we must therefore calculate the area of overlap of two circles of differing radii.^{12,13} As the system has circular symmetry, $C_W(l)$ is a function of a radial spatial frequency l only, where l is normalized so that the cutoff of the objective pupil corresponding to a spatial frequency $\sin \alpha_o / \lambda$, where λ is the wavelength, occurs at $l = 1$. The result depends on whether the two circles intersect. We obtain

$$C_W(l) = \pi S^2, \quad 0 \leq l \leq 1 - S, \quad (2)$$

$$C_W(l) = S^2 \arccos\left(\frac{l^2 + S^2 - 1}{2lS}\right) - \left(\frac{l^2 + S^2 - 1}{2l}\right) \times \left[S^2 - \left(\frac{l^2 + S^2 - 1}{2l}\right)^{1/2}\right] + \arccos\left(\frac{l^2 - S^2 + 1}{2l}\right) - \left(\frac{l^2 - S^2 + 1}{2l}\right) \times \left[1 - \left(\frac{l^2 - S^2 + 1}{2l}\right)^{1/2}\right], \quad 1 - S \leq l \leq 1 + S. \quad (3)$$

This is illustrated, after normalization, in Fig. 1. The bandwidth of the system increases with S .

3. DEFOCUSED WEAK-OBJECT TRANSFER FUNCTION

For the defocused case, we must perform the integration over the effective pupil,

$$P(\rho) = \exp(-\frac{1}{2} i u \rho^2), \quad \rho < 1, \quad (4)$$

where u is the axial optical coordinate,^{14,15}

$$u = 2W_{20} = 4kz \sin^2(\alpha_o/2), \quad (5)$$

$k = 2\pi/\lambda$, and W_{20} is the Seidel aberration coefficient for defocus. Then, for the in-focus case, $u = 0$ and

$$C_W(l; 0) = \int_{-(s+l/2)}^{s-l/2} 2 \left[S^2 - \left(\frac{l}{2} + x \right)^2 \right]^{1/2} dx = \pi S^2 \quad (6)$$

for $0 \leq l \leq 1 - S$, as before.

For the defocused case,

$$C_W(l; u) = \int_{-(S+l/2)}^{S-l/2} 2 \left[S^2 - \left(\frac{l}{2} + x \right)^2 \right]^{1/2} \times \exp(iulx) dx, \quad 0 \leq l \leq 1 - S, \quad (7)$$

$$C_W(l; u) = \int_{-(1-S^2)/2l}^{S-l/2} 2 \left[S^2 - \left(\frac{l}{2} + x \right)^2 \right]^{1/2} \times \exp(iulx) dx + \int_{l/2-1}^{-(1-S^2)/2l} 2 \left[1 - \left(\frac{l}{2} - x \right)^2 \right]^{1/2} \times \exp(iulx) dx, \quad 1 - S \leq l \leq 1 + S. \quad (8)$$

The WOTF is complex, and the real and imaginary parts, calculated from Eqs. (7) and (8) are shown in Fig. 2. For

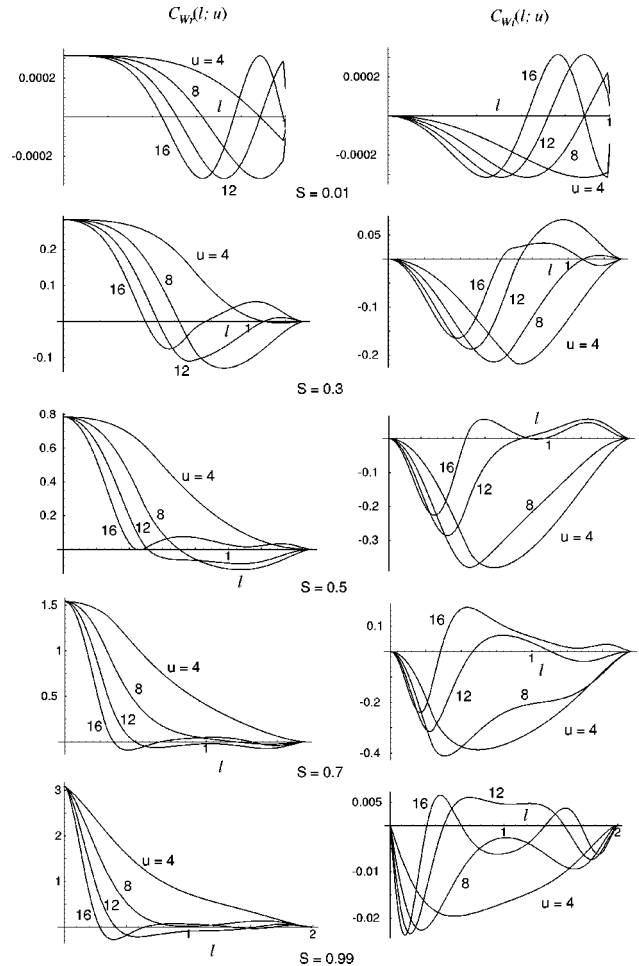


Fig. 2. Defocused WOTF for a partially coherent bright-field microscope with coherence parameter S .

small S , the real and imaginary parts of the WOTF are given by the cosine and sine, respectively, of a constant multiplied by l^2 and fall rapidly to zero at the cutoff. The behavior of the real part for S close to unity agrees well with the results for the OTF given by Hopkins.¹ For larger values of u the WOTF oscillates, and the real part can go negative. Some spatial-frequency components are thus added in counter phase, resulting in image artifacts.

4. WEAK DEFOCUS AND PHASE RETRIEVAL

For the case when the defocus is small, the exponential functions in Eqs. (7) and (8) can be expanded to the first order by the binomial theorem, so that the real part C_{Wr} is given by Eqs. (2) and (3), and the imaginary part C_{Wi} is given by the analytic expressions

$$C_{Wi} = -\frac{1}{2}\pi ul^2 S^2, \quad 0 \leq l \leq 1 - S, \quad (9)$$

$$\begin{aligned} C_{Wi} = & -\frac{ul^2}{2} \left[S^2 \arccos\left(\frac{l^2 + S^2 - 1}{2lS}\right) \right. \\ & \left. - \arccos\left(\frac{l^2 - S^2 + 1}{2l}\right) \right] \\ & - \frac{u}{6l} \left[\left[S^2 - \left(\frac{l^2 + S^2 - 1}{2l}\right)^2 \right]^{1/2} \right. \\ & \times \left[(1 - S^2)^2 - \frac{l^2}{2}(1 + l^2 + 7S^2) \right] \\ & \left. - \left[1 - \left(\frac{l^2 - S^2 + 1}{2l}\right)^2 \right]^{1/2} \right. \\ & \left. \times \left[(1 - S^2)^2 - \frac{l^2}{2}(7 + l^2 + S^2) \right] \right], \\ & 1 - S \leq l \leq 1 + S. \quad (10) \end{aligned}$$

These are proportional to the defocus and are plotted in Fig. 3. The curves all show a monotonic increase with l to a maximum value and then fall monotonically. The maximum value is maximized for a value of S of ~ 0.6 . The behavior agrees well with that shown in Fig. 2 for $u = 4$, except for the smallest values of S .

It is seen from Eq. (9) that the imaginary part of the WOTF is proportional to the area of the condenser pupil; therefore we can normalize so that the power incident on the specimen is kept constant. The WOTF is also proportional to l^2 for $0 \leq l \leq 1 - S$, so that if the image is processed by dividing by a simple filter

$$F(l) = -\frac{1}{2}\pi ul^2 S^2, \quad (11)$$

the WOTF after restoration (the restored WOTF, C_{Wir}) is then

$$C_{Wir} = 1, \quad 0 \leq l \leq 1 - S, \quad (12)$$

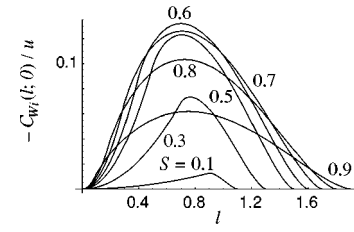


Fig. 3. (Negative of the) imaginary part of the WOTF for a weakly defocused, partially coherent bright-field microscope with coherence parameter S .

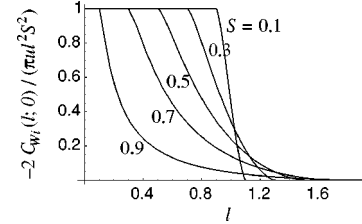


Fig. 4. (Negative of the) imaginary part of the WOTF for a weakly defocused, partially coherent bright-field microscope with coherence parameter S , after restoration with an inverse Laplacian filter.

$$\begin{aligned} C_{Wir} = & \frac{1}{2\pi S^2} \left[S^2 \arccos\left(\frac{l^2 + S^2 - 1}{2lS}\right) \right. \\ & \left. - \arccos\left(\frac{l^2 - S^2 + 1}{2l}\right) \right] \\ & + \frac{1}{6l^3 \pi S^2} \left[\left[S^2 - \left(\frac{l^2 + S^2 - 1}{2l}\right)^2 \right]^{1/2} \right. \\ & \times \left[(1 - S^2)^2 - \frac{l^2}{2}(1 + l^2 + 7S^2) \right] \\ & \left. - \left[1 - \left(\frac{l^2 - S^2 + 1}{2l}\right)^2 \right]^{1/2} \right. \\ & \left. \times \left[(1 - S^2)^2 - \frac{l^2}{2}(7 + l^2 + S^2) \right] \right], \\ & 1 - S \leq l \leq 1 + S. \quad (13) \end{aligned}$$

The filter can be recognized as an inverse Laplacian filter.¹¹ The restored WOTF is shown in Fig. 4. The response is linear in phase up to a normalized frequency $1 - S$ and then rolls off to a cutoff at $1 + S$. If an object with arbitrary weak phase and amplitude is imaged with equal and opposite small defoci $u, -u$, then addition of the two images gives a pure amplitude-contrast image, i.e., an image of the transmittance of the object, while subtraction of the two images gives a phase-contrast image that is also proportional to the transmittance of the object. This was pointed out by Barone-Nugent *et al.*⁸ and results from the fact that the pupil function and hence the WOTF is a Hermitian function of defocus.⁴ The phase-contrast image can be restored by using the inverse Laplacian filter and is linear in phase to the first order for spatial frequencies in the range $0 \leq l \leq 1 - S$. A pure phase image can be extracted by dividing the phase-contrast image by the amplitude-contrast image. How-

ever, as the transfer functions for the real and imaginary parts are different (compare Figs. 1 and 4), the value of the phase will be distorted for $l > 1 - S$ unless one of the images is filtered.

5. DISCUSSION

It is well known that the WOTF can be used to extract the quantitative phase and amplitude of a weak object provided that the amplitude and phase changes are small enough for the bilinear terms to be neglected. The defocused weak-object transfer function for a partially coherent optical system has been calculated directly by an integral over the pupil functions of the condenser and the objective lenses. For weak defocus, an analytic expression is given for the defocused WOTF. For a strong phase object this approach breaks down, although if the object varies slowly over dimensions of the order of the point-spread function, a different simplified treatment based on ray tracing (the phase-gradient transfer function¹⁶) can be used. This is beyond the scope of the present paper. For a general strong phase object, these simple approaches fail, and the full partially coherent transmission cross coefficient must be used to calculate the intensity in the image.

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